

Lab 6 Solution

```
library(tidyverse)
library(sf)
```

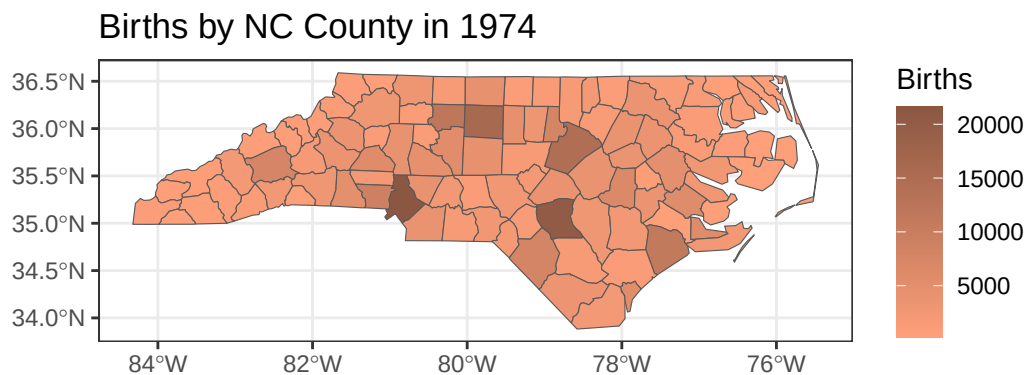
Exercise 1

a.

```
nc <- st_read(system.file("shape/nc.shp", package = "sf"),
               quiet = TRUE)
```

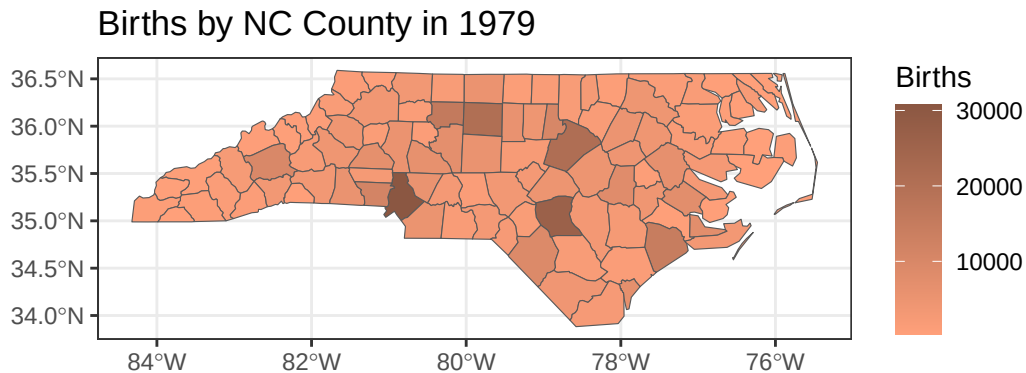
b.

```
ggplot(nc) +
  geom_sf(aes(fill = BIR74)) +
  scale_fill_gradient(low = "lightsalmon", high = "lightsalmon4") +
  theme_bw() +
  labs(title = "Births by NC County in 1974",
       fill = "Births")
```



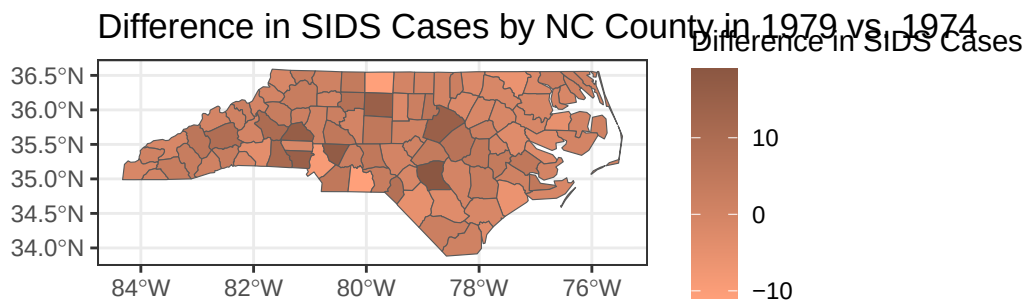
Exercise 2

```
ggplot(nc) +  
  geom_sf(aes(fill = BIR79)) +  
  scale_fill_gradient(low = "lightsalmon", high = "lightsalmon4") +  
  theme_bw() +  
  labs(title = "Births by NC County in 1979",  
        fill = "Births")
```



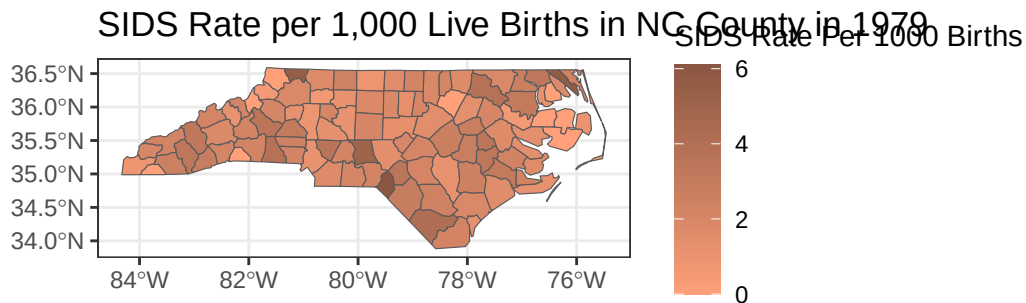
Exercise 3

```
ggplot(nc) +  
  geom_sf(aes(fill = SID79-SID74)) +  
  scale_fill_gradient(low = "lightsalmon", high = "lightsalmon4") +  
  theme_bw() +  
  labs(title = "Difference in SIDS Cases by NC County in 1979 vs. 1974",  
        fill = "Difference in SIDS Cases")
```



Exercise 4

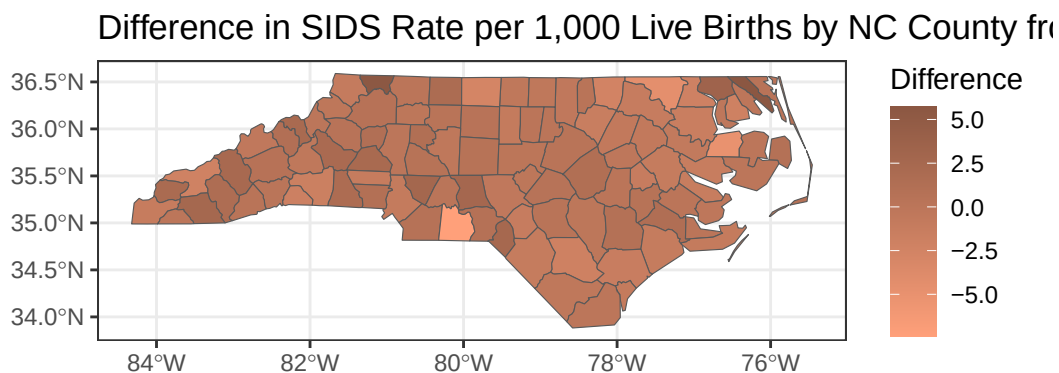
```
ggplot(nc) +
  geom_sf(aes(fill = SID79/BIR79*1000)) +
  scale_fill_gradient(low = "lightsalmon", high = "lightsalmon4") +
  theme_bw() +
  labs(title = "SIDS Rate per 1,000 Live Births in NC County in 1979",
       fill = "SIDS Rate Per 1000 Births")
```



Exercise 5

a.

```
ggplot(nc) +
  geom_sf(aes(fill = SID79 / BIR79 *1000- SID74 / BIR74 *1000)) +
  scale_fill_gradient(low = "lightsalmon",
                     high = "lightsalmon4") +
  theme_bw() + labs(fill = "Difference",
                    title = "Difference in SIDS Rate per 1,000 Live Births by NC County from 1974 to 1979")
```



b. Yes, a few counties with a decrease. South central and Northeast counties.

Exercise 6

```
nc |>
  arrange(desc(SID79)) |>
  select(NAME, SID79) |>
  slice(1:5) |>
  st_drop_geometry()
```

	NAME	SID79
1	Cumberland	57
2	Guilford	38
3	Mecklenburg	35
4	Wake	31
5	Gaston	26

b.

```
nc |>
  mutate(rate_79 = SID79 / BIR79 * 1000) |>
  arrange(desc(rate_79)) |>
  select(NAME, rate_79) |>
  slice(1:5) |>
  st_drop_geometry()
```

	NAME	rate_79
1	Scotland	6.113871
2	Camden	5.714286
3	Alleghany	5.535055
4	Montgomery	5.006258
5	Columbus	4.102317

c.

```
nc |>
  mutate(increase = SID79-SID74) |>
  arrange(desc(increase)) |>
  select(NAME, increase) |>
  slice(1:5) |>
  st_drop_geometry()
```

	NAME	increase
1	Cumberland	19
2	Cabarrus	17
3	Catawba	16
4	Guilford	15
5	Wake	15

Exercise 6

If I were a biometrician working for North Carolina in 1980, I would focus on counties which saw the largest “regression” from 1974 to 1979, such as Cumberland and Cabarrus counties – those counties colored in most dark in Question 3, or included in the final table in Question 6. The reason I would focus on those counties is something is probably happening related to the increase in SIDS cases, so by focusing resources there NC might be able to arrest this trend before it continues.